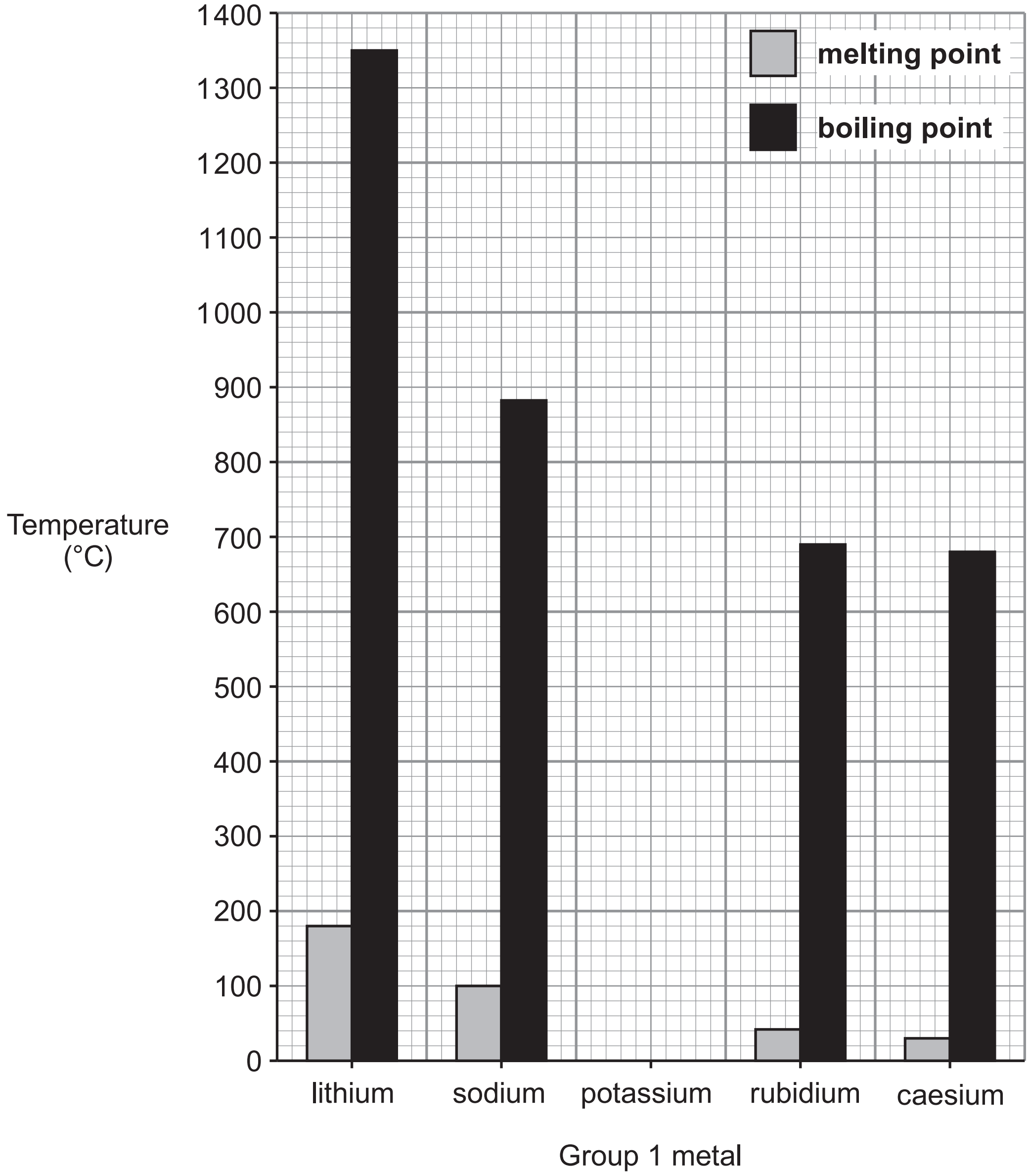
3. *(a)* The chart below shows the melting points and boiling points of some Group 1 metals. The data for potassium is missing.



Tick (✓) the box that shows the melting point and boiling point values for potassium which fit the trends shown in the chart. [1]

melting point = 40 °C
boiling point = 600 °C ☐

melting point = 60 °C
boiling point = 780 °C ☐

melting point = 20 °C
boiling point = 600 °C ☐

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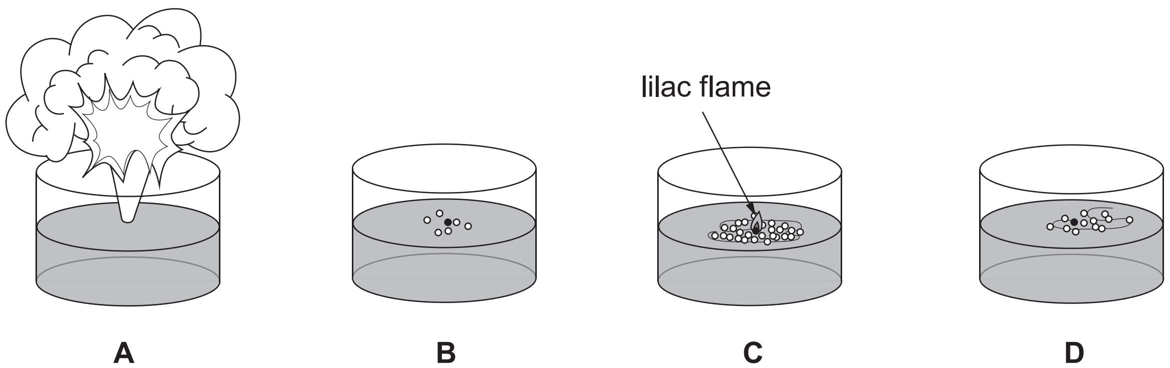
(b) The diagrams show lithium, sodium, potassium and rubidium, **but not necessarily in that order**, reacting separately with cold water.



A	B	C	D
Violent reaction with sparks spitting out of the trough	Gentle fizzing	Metal ball melts, moves around on the surface and burns	Metal ball fizzes and moves around on the surface

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(i) Use the information in the diagrams to give the **letter** which represents: [2]

lithium ....

sodium ....

potassium ....

rubidium ....

(ii) Caesium lies below rubidium in Group 1. Suggest how its reaction with water would be different from those above. [1]

....

(iii) Describe **one** safety precaution taken when adding a Group 1 metal to water. [1]

....

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(c) Group 1 metals react with oxygen to form metal oxides.

Sodium oxide contains the ions Na^+ and O^{2-}.

Underline the correct formula of sodium oxide. [1]

2NaO **NaO_2** **Na_2O** **Na^2O**

(d) Group 1 metals react with chlorine, Cl_2, to form metal chlorides.

Give the **letter** for the balanced symbol equation for the reaction between sodium and chlorine. [1]

A $Na + Cl \longrightarrow NaCl$

B $2Na + 2Cl \longrightarrow 2NaCl$

C $2Na + Cl_2 \longrightarrow 2NaCl$

D $Na + Cl_2 \longrightarrow NaCl_2$

Letter ....

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